Black–Scholes Equation with Gamma Constraint

Houssem Bouabid & Tingjia ZHANG

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Supervisor : Alessandro Zilio

Introduction

We consider an extension of the Black–Scholes framework in which the replication strategy is subject to market constraints, modeled via an upper bound on the option's curvature (Gamma). Let S_t denote the risky asset and S_t^0 the bank account :

$$dS_t = S_t\big(rdt + \sigma(t, S_t)\, dB_t\big), \qquad dS_t^0 = rS_t^0\, dt,$$

where $r > 0$ is constant.

In the classical complete-market setting, the hedging strategy holds $Y_t = \partial_s u(t, S_t)$, where u solves the linear Black–Scholes PDE. To account for market constraints that restrict how aggressively a hedger can rebalance risk exposure, we impose a Gamma constraint on the option price $v(t, s)$:

$$s\, v_{ss}(t, s) \leq \Gamma, \qquad \Gamma > 0. \tag{1}$$

This bound limits the local convexity of the price function, reflecting practical restrictions on rapid inventory adjustments. As a consequence, the classical perfect-replication argument no longer applies and the valuation problem becomes nonlinear.

Nonlinear variational inequality

Under the constraint (1), the price v is characterized by the following variational inequality :

$$\begin{cases} \min\big(-\partial_t v - \frac{1}{2}\sigma^2(t, s)s^2\, \partial_{ss} v - rs\, \partial_s v + rv,\ \Gamma - s\, \partial_{ss} v\big) = 0, \\ v(T, s) = \hat{g}(s). \end{cases} \tag{2}$$

The first term corresponds to the standard Black–Scholes operator, while the second term enforces the curvature cap. Informally :

— if $sv_{ss}(t, s) < \Gamma$, the constraint is inactive and v satisfies the Black–Scholes PDE locally ;

— if $sv_{ss}(t, s) = \Gamma$, the constraint binds and the curvature is saturated.

Gamma-constrained terminal condition

The terminal condition is not the raw payoff g, but its Γ-constrained envelope $\hat{g}$, defined (for $s > 0$) as the smallest function above g whose curvature is capped by Γ :

$$\hat{g}(s) \geq g(s), \tag{3}$$

$$s\, \hat{g}''(s) \leq \Gamma. \tag{4}$$

It can be compactly characterized by the stationary variational inequality

$$\min\big(\hat{g}(s) - g(s),\ \Gamma - s\, \hat{g}''(s)\big) = 0, \qquad s > 0. \tag{5}$$

Q1 : Gamma-constrained envelope for a European put

Problem setup and scaling

We take the put payoff

$$g(S) = (K - S)^+,$$

and construct its Gamma-constrained envelope $\hat{g}$, defined as the solution of (5).

To remove dimensional issues in logarithms and reduce parameters, we introduce

$$x = \frac{S}{K}, \qquad \varphi(x) = \frac{\hat{g}(S)}{K}.$$

Then the payoff becomes

$$\varphi_{\text{payoff}}(x) = (1 - x)^+,$$

and, using $\hat{g}''(S) = \varphi''(x)/K$, the constraint (4) becomes

$$x\,\varphi''(x) \le \Gamma. \tag{6}$$

In this section we derive the explicit free-boundary form of φ and provide a Newton method that recovers the same parameters.

Free-boundary structure and explicit solution

For a put payoff, both sides are linear : $g(S) = K - S$ for $S < K$ and $g(S) = 0$ for $S \ge K$. Hence $g''(S) = 0$ on each side, so the Gamma cap is automatically satisfied there. The constraint only becomes relevant in the intermediate transition region, leading to two free boundaries $0 < \alpha < 1 < \beta$ such that

$$\varphi(x) = \begin{cases} 1 - x, & x \le \alpha, \\ \Gamma(x \ln x - x) + C_1 x + C_2, & \alpha < x < \beta, \\ 0, & x \ge \beta, \end{cases} \tag{7}$$

where the middle branch corresponds to the active constraint (4).

The four unknown parameters $(\alpha, \beta, C_1, C_2)$ are determined by *smooth pasting* at the free boundaries :

$$\varphi(\alpha) = 1 - \alpha, \qquad \varphi'(\alpha) = -1, \qquad \varphi(\beta) = 0, \qquad \varphi'(\beta) = 0. \tag{8}$$

Proposition 1 (Closed-form parameters)**.** *The smooth pasting system* (8) *admits the explicit solution*

$$\beta = \frac{1}{\Gamma\big(1 - e^{-1/\Gamma}\big)}, \qquad \alpha = \beta e^{-1/\Gamma} = \frac{1}{\Gamma\big(e^{1/\Gamma} - 1\big)},$$

and

$$C_1 = -\Gamma \ln \beta, \qquad C_2 = \Gamma\beta.$$

Démonstration. From $\varphi'(\beta) = 0$ we get $C_1 = -\Gamma \ln \beta$. Then $\varphi(\beta) = 0$ yields

$$0 = \Gamma(\beta \ln \beta - \beta) + C_1\beta + C_2 = \Gamma(\beta \ln \beta - \beta) - \Gamma\beta \ln \beta + C_2 = -\Gamma\beta + C_2,$$

hence $C_2 = \Gamma\beta$. Next $\varphi'(\alpha) = -1$ implies

$$\Gamma \ln \alpha + C_1 = -1 \quad \Longrightarrow \quad \Gamma \ln(\alpha/\beta) = -1 \quad \Longrightarrow \quad \alpha = \beta e^{-1/\Gamma}.$$

Finally, $\varphi(\alpha) = 1 - \alpha$ gives

$$1 - \alpha = \Gamma\big(\alpha \ln(\alpha/\beta) - \alpha + \beta\big) = \Gamma\big(-\alpha/\Gamma - \alpha + \beta\big) = \Gamma\beta - (\Gamma + 1)\alpha,$$

which reduces to $\Gamma(\beta - \alpha) = 1$. Substituting $\alpha = \beta e^{-1/\Gamma}$ yields $\beta(1 - e^{-1/\Gamma}) = 1/\Gamma$, hence the stated formulas. $\square$

Proposition 2 (Numerical verification - Newton meth)**.** *Although Q1 admits a closed form, we present a Newton solver that will be useful for payoffs without explicit solutions and serves as a numerical verification.*

Let $\theta = (\alpha, \beta, C_1, C_2)$ *and define, using the active-region expressions*

$$\varphi(x) = \Gamma(x \ln x - x) + C_1 x + C_2, \qquad \varphi'(x) = \Gamma \ln x + C_1,$$

the residual vector $F(\theta) \in \mathbb{R}^4$ *:*

$$F_1(\theta) = \varphi(\alpha) - (1-\alpha), \quad F_2(\theta) = \varphi'(\alpha) + 1, \quad F_3(\theta) = \varphi(\beta), \quad F_4(\theta) = \varphi'(\beta).$$

Newton's method iterates

$$\theta^{(k+1)} = \theta^{(k)} - J_F(\theta^{(k)})^{-1} F(\theta^{(k)}),$$

until $\|F(\theta^{(k)})\|$ *is below tolerance.*

Numerical values for $\Gamma = 1$

We now illustrate the closed-form solution of Proposition 1 and verify it numerically using the Newton solver of Proposition 2 for $\Gamma = 1$.

Parameter	Newton solution	Closed-form
α	0.5819767068693265	0.5819767068693265
β	1.5819767068693265	1.5819767068693265
C_1	−0.4586751453870819	−0.4586751453870819
C_2	1.5819767068693265	1.5819767068693265

TABLE 1 – Newton verification of the free-boundary parameters for $\Gamma = 1$.

In our experiments, convergence is rapid and the recovered parameters are nearly identical to the closed-form values.

Shape of the envelope

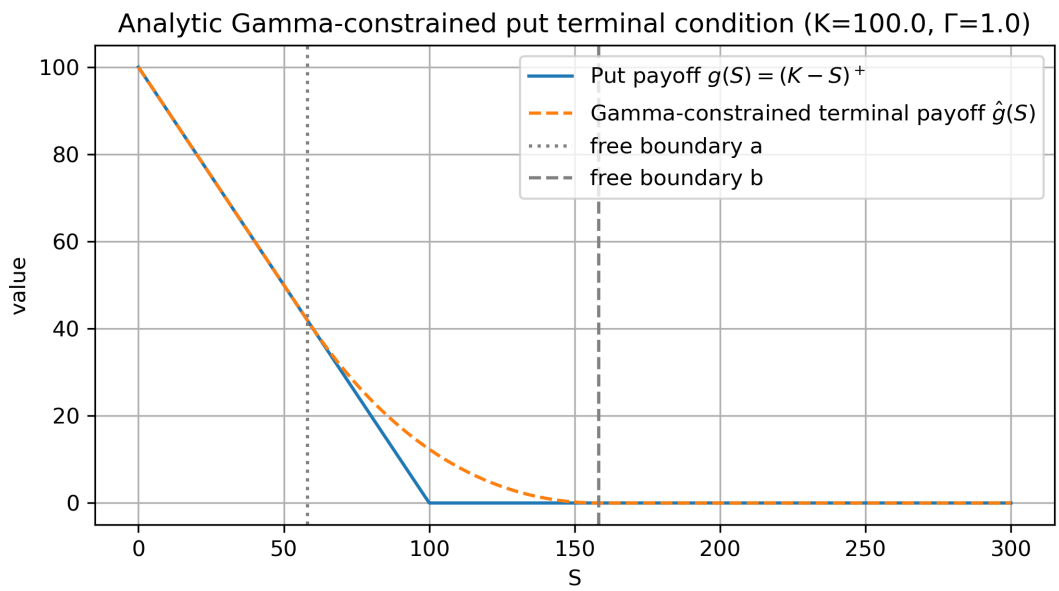


FIGURE 1 – Gamma-constrained put envelope $\hat{g}$.

Q2 : Numerical solution of the Gamma-constrained Black–Scholes problem

Time-to-maturity formulation

We work with time-to-maturity

$$\tau = T - t, \qquad u(\tau, s) = v(T - \tau, s), \qquad 0 \le \tau \le T,$$

so that (2) becomes

$$\min\Big(u_\tau - \tfrac{1}{2}\sigma^2 s^2 u_{ss} - rsu_s + ru,\ \Gamma - s\,u_{ss}\Big) = 0, \qquad u(0, s) = \hat{g}(s). \tag{9}$$

Grid, initial and boundary conditions

We truncate to $s \in [0, S_{\max}]$ and use uniform grids

$$s_i = i\Delta s\ (i = 0, \dots, M), \qquad \tau_n = n\Delta\tau\ (n = 0, \dots, N),$$

with $\Delta s = S_{\max}/M$, $\Delta\tau = T/N$, and $u_i^n \approx u(\tau_n, s_i)$. The initial condition is always $u_i^0 = \hat{g}(s_i)$. We impose Dirichlet conditions at $s = 0$ and $s = S_{\max}$ (Case 1 : $u(\tau, 0) = Ke^{-r\tau}$, Case 2 : $u(\tau, 0) = 0$, and $u(\tau, S_{\max}) = 0$).

PDE branch : θ-scheme (Euler / CN / mixed)

Let $\mathcal{L}u = \frac{1}{2}\sigma^2 s^2 u_{ss} + rsu_s - ru$ and $\mathcal{L}_h$ its standard centered FD discretization. On interior nodes $1 \le i \le M - 1$, the PDE branch is discretized by the θ-scheme

$$\frac{u_i^{n+1} - u_i^n}{\Delta\tau} - \Big(\theta_n(\mathcal{L}_h u^{n+1})_i + (1 - \theta_n)(\mathcal{L}_h u^n)_i\Big) = 0. \tag{10}$$

We use

$$\theta_n = \begin{cases} 1 & \text{(implicit Euler)}, \\ \frac{1}{2} & \text{(Crank–Nicolson)}, \\ 1 \text{ for } n < n_R,\ \frac{1}{2} \text{ for } n \ge n_R & \text{(mixed / Rannacher, typically } n_R = 2). \end{cases}$$

For a fixed n (hence fixed θ_n), (10) yields a tridiagonal linear system

$$B_0^{(n)} u^{n+1} = b_0^{(n)}(u^n),$$

where boundary values are incorporated into the right-hand side.

Howard form and policy iteration

The constraint branch is discretized by

$$(B_1 u^{n+1} - b_1)_i := \Gamma - s_i(D_2 u^{n+1})_i, \qquad (b_1)_i = \Gamma, \qquad 1 \le i \le M - 1,$$

so that at each time step $n \to n + 1$ the VI (9) can be written in the *Howard canonical form*

$$\min\Big(B_0^{(n)} u^{n+1} - b_0^{(n)}(u^n),\ B_1 u^{n+1} - b_1\Big) = 0, \tag{11}$$

where the minimum is taken componentwise.

Introduce a policy $\alpha \in \{0, 1\}^{M-1}$ and define $B(\alpha), b(\alpha)$ row-wise by selecting, at each node i, the PDE row ($\alpha_i = 0$) or the constraint row ($\alpha_i = 1$). Then (11) is equivalent to

$$\min_{\alpha \in \{0,1\}^{M-1}} \big(B(\alpha) u^{n+1} - b(\alpha)\big) = 0.$$

Howard's algorithm iterates :

1. **Policy update :**

$$\alpha_i^{(k+1)} = \arg\min_{a\in\{0,1\}} \big((B_a x^{(k)} - b_a)_i\big), \qquad x^{(k)} \approx u^{n+1}.$$

2. **Policy evaluation :** solve the linear system

$$B(\alpha^{(k+1)})x^{(k+1)} = b(\alpha^{(k+1)}).$$

3. Stop when the policy stabilizes.

Since each policy evaluation solves a tridiagonal system in one dimension, the method is efficient and robust.

Numerical experiments

We use the parameters specified in the project statement :

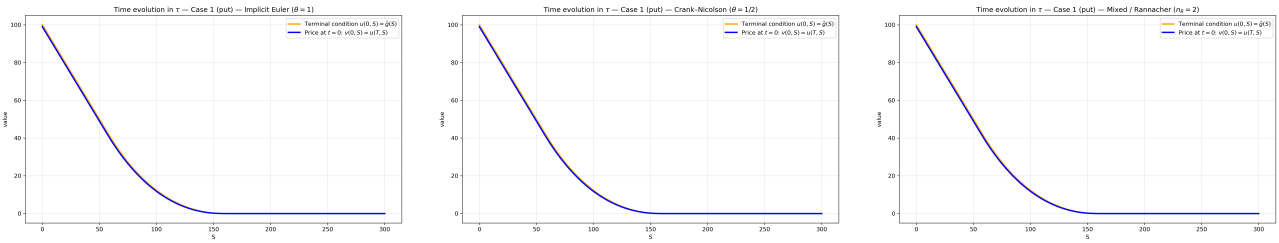
$$K = 100, \qquad \sigma = 0.1, \qquad r = 0.1, \qquad T = 0.1, \qquad \Gamma = 1.$$

We consider two terminal payoffs :

1. **Case 1 (put) :** $g(s) = (K - s)^+$, with analytic $\hat{g}_1$ from Q1 ;
2. **Case 2 (triangle) :** $g(s) = (K - 2|s - K|)^+$, with $\hat{g}_2$ computed numerically.

We compare three time-stepping choices : pure implicit Euler, pure Crank–Nicolson, and the mixed Rannacher strategy. For the mixed scheme we take $n_R = 2$, i.e. two implicit-Euler start-up steps followed by Crank–Nicolson.

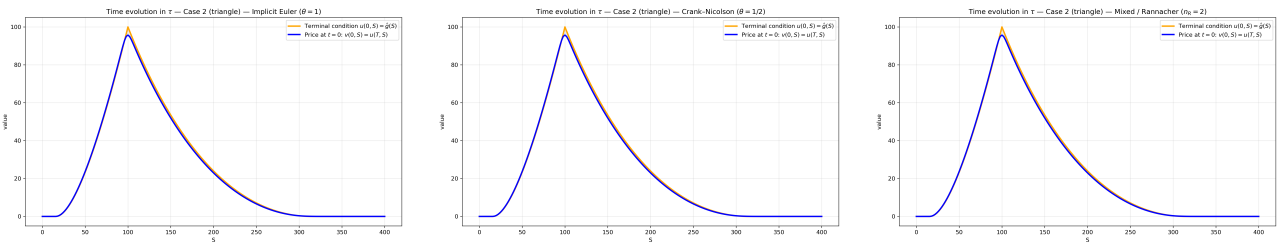
Case 1 : put payoff — time evolution (Euler / CN / Mixed)



(a) Implicit Euler ($\theta = 1$) (b) Crank–Nicolson ($\theta = \frac{1}{2}$) (c) Mixed / Rannacher ($n_R = 2$)

FIGURE 2 – Case 1 (put) : time evolution of $u(\tau, S)$ for three time-stepping choices.

Case 2 : triangle payoff — time evolution (Euler / CN / Mixed)



(a) Implicit Euler ($\theta = 1$) (b) Crank–Nicolson ($\theta = \frac{1}{2}$) (c) Mixed / Rannacher ($n_R = 2$)

FIGURE 3 – Case 2 (triangle) : time evolution of $u(\tau, S)$ for three time-stepping choices.

We observe that the three schemes yield nearly indistinguishable results, and the resulting time evolution is fully consistent with the expected option-price dynamics. In the next step, we compare our numerical solutions with the exact Black–Scholes benchmark.

Q3 : Comparison with the classical Black–Scholes model

Comparison setup and benchmark

The objective of this section is to quantify the impact of the Gamma constraint on option values. To ensure a fair comparison, we compute :

— the **Gamma-constrained** price $v(t, S)$ as the solution of (2) ;

— the **classical Black–Scholes** price $u(t, S)$ as the risk-neutral value with the same terminal condition $\hat{g}$:

$$u(t, S) = e^{-r(T-t)}\, \mathbb{E}^{\mathbb{Q}}[\hat{g}(S_T) \mid S_t = S]\,, \qquad S_T = S \exp\Big((r - \tfrac{1}{2}\sigma^2)(T-t) + \sigma\sqrt{T-t}\, Z\Big), \quad (12)$$

where $Z \sim \mathcal{N}(0, 1)$.

Discussion for expected behavior

From an economic viewpoint, restricting admissible hedging strategies typically increases super-replication costs, hence one expects a nonnegative *Gamma-constraint premium* $v-u$ in regions where the constraint is active. However, in the discrete implementation, the pointwise sign of $v - u$ can exhibit local oscillations, especially near non-smooth features of $\hat{g}$ and near the boundary of the active set.

Numerical implementation

We compute v using the Howard finite-difference solver described in Q2. For u, we use the semi-closed-form representation (12) : on each time level, $u(\tau, \cdot)$ is obtained by evaluating expectations of $\hat{g}(S_\tau)$ using truncated lognormal moments. Both models use the same parameters $(K, \sigma, r, T, \Gamma)$ and the same terminal condition $\hat{g}$.

Results : Case 1 (put)

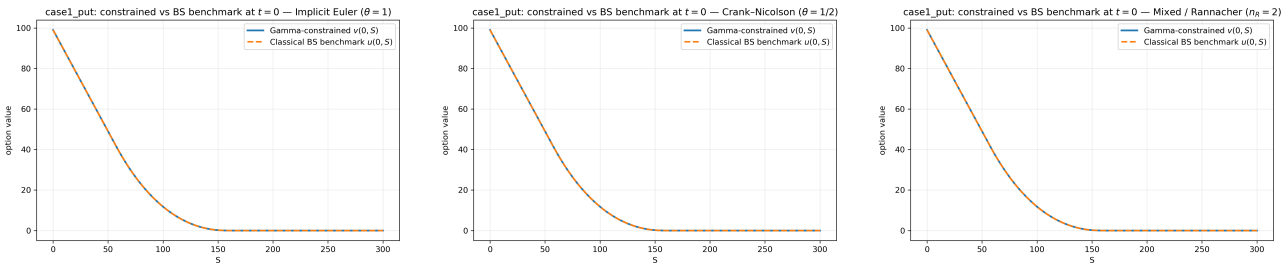
Figures 4–5 report the $t = 0$ comparison for the put experiment. Across all three schemes, the difference profile $v(0, S) - u(0, S)$ is small (on the order of 10^{-4}–10^{-3}) and localized near the region where the Gamma cap is active. Quantitatively :

— **Implicit Euler :** $\min(v - u) \approx -4.06 \times 10^{-4}, \quad \max(v - u) \approx 9.04 \times 10^{-4}$;

— **Crank–Nicolson :** $\min(v - u) \approx -3.72 \times 10^{-4}, \quad \max(v - u) \approx 8.53 \times 10^{-4}$;

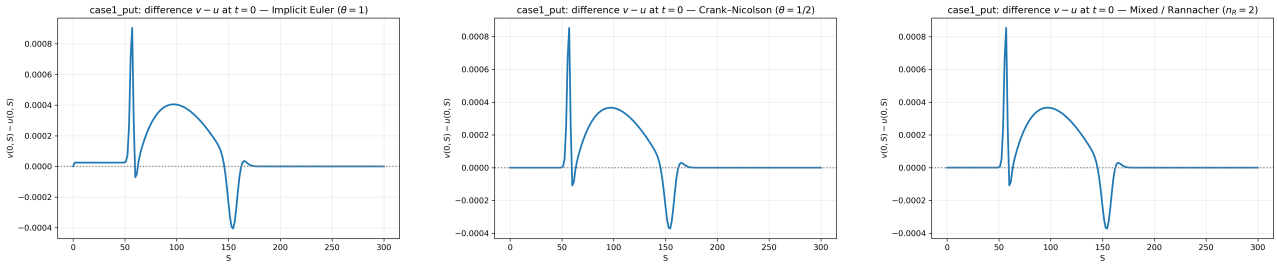
— **Mixed :** $\min(v - u) \approx -3.72 \times 10^{-4}, \quad \max(v - u) \approx 8.53 \times 10^{-4}$.

The negative undershoot is mild and comparable across schemes, consistent with discretization and truncation effects rather than a systematic reversal.



(a) Implicit Euler (b) Crank–Nicolson (c) Mixed / Rannacher

FIGURE 4 – Case 1 (put) : Gamma-constrained price versus classical BS benchmark at $t = 0$.



(a) Implicit Euler (b) Crank–Nicolson (c) Mixed / Rannacher

FIGURE 5 – Case 1 (put) : difference $v(0,S) - u(0,S)$ under the three schemes.

Results : Case 2 (triangle payoff)

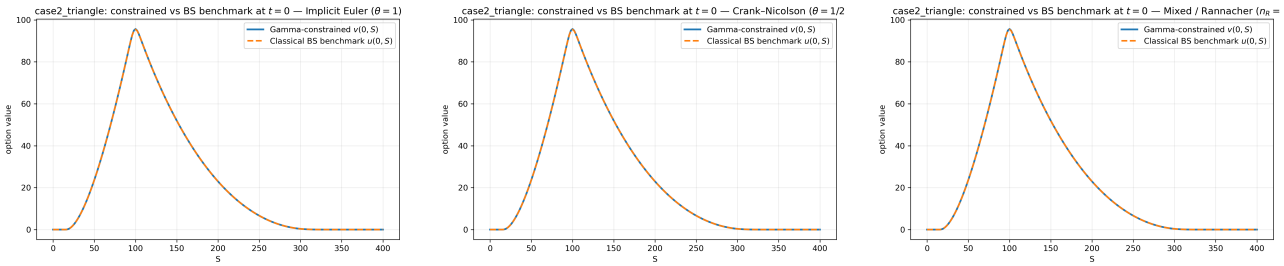
For the triangle payoff, the envelope $\hat{g}$ exhibits sharper features and a wider effective support, so the constrained dynamics is more sensitive to active-set switching and domain truncation. As a result, the magnitude of $v - u$ is much larger than in Case 1. Quantitatively :

— **Implicit Euler :** $\min(v-u) \approx -3.04 \times 10^{-2}, \quad \max(v-u) \approx 2.32 \times 10^{-1}$;

— **Crank–Nicolson :** $\min(v-u) \approx -2.82 \times 10^{-2}, \quad \max(v-u) \approx 2.30 \times 10^{-1}$;

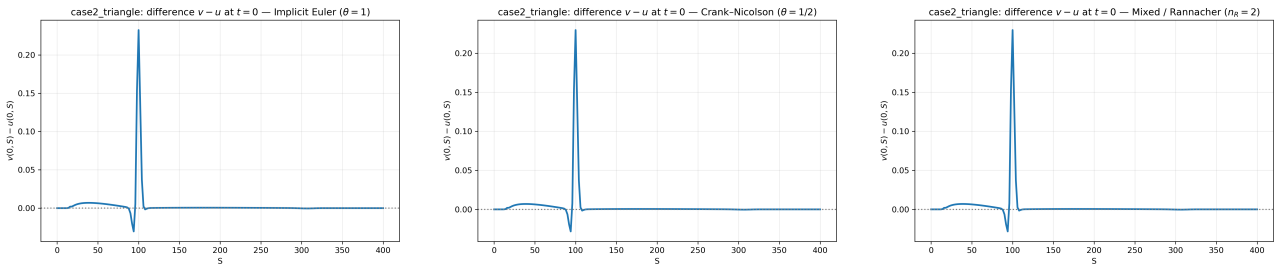
— **Mixed :** $\min(v-u) \approx -2.83 \times 10^{-2}, \quad \max(v-u) \approx 2.30 \times 10^{-1}$.

The discrepancy is concentrated around the region where the envelope is most curved and where the Gamma cap binds along the evolution.



(a) Implicit Euler (b) Crank–Nicolson (c) Mixed / Rannacher

FIGURE 6 – Case 2 (triangle) : Gamma-constrained price versus classical BS benchmark at $t = 0$.



(a) Implicit Euler (b) Crank–Nicolson (c) Mixed / Rannacher

FIGURE 7 – Case 2 (triangle) : difference $v(0,S) - u(0,S)$ under the three schemes.

Discussion

Across both experiments, the Gamma constraint produces a clear quantitative impact concentrated in regions where curvature demand is high. Case 1 shows a small discrepancy (typically below 10^{-3} in absolute value), whereas Case 2 leads to a much larger discrepancy, consistent with a sharper terminal envelope and a more frequently binding Gamma cap. The observed negative values of $v - u$ should be

interpreted as numerical artifacts induced by discretization, finite-domain truncation, and active-set switching, rather than as a fundamental economic reversal of the constraint effect.

Q4 : Numerical solution of the Gamma-constrained Black–Scholes problem with transaction costs (Leland, 1985)

Motivation for the Leland model

The Black–Scholes framework relies on the assumption of frictionless markets, where transaction costs are zero and hedging can be performed continuously. In practice, these assumptions are unrealistic : any dynamic hedging strategy generates proportional transaction fees, and perfect replication is no longer achievable.

The Leland (1985) model extends the Black–Scholes framework by explicitly incorporating proportional transaction costs. It shows that the cumulative effect of these costs can be captured through a modification of the volatility parameter, leading to a pricing equation that preserves the Black–Scholes structure while accounting for hedging expenses.

Leland pricing equation

Let $v(t,s)$ denote the price of a European contingent claim with maturity T and payoff $\hat{g}(s)$. In the Leland framework, transaction costs generated by discrete delta-hedging are incorporated through an effective modification of the diffusion term of the pricing equation. The option price satisfies the following nonlinear partial differential equation :

$$\min\left[-\frac{\partial v}{\partial t}-\frac{1}{2}\,\sigma^2(t,s)\,s^2\left(\frac{\partial^2 v}{\partial s^2}+\mathrm{Le}\left|\frac{\partial^2 v}{\partial s^2}\right|\right)-rs\frac{\partial v}{\partial s}+rv,\ \Gamma-s\frac{\partial^2 v}{\partial s^2}\right]=0, \tag{13}$$

with terminal condition

$$v(T,s)=\hat{g}(s).$$

The additional nonlinear term involving the absolute value of the second derivative models the cumulative effect of proportional transaction costs generated by discrete rebalancing of the hedging portfolio. The dimensionless coefficient Le is called the *Leland number* and measures the intensity of transaction costs. It is assumed to be small.

When $\mathrm{Le}=0$, equation (13) reduces to the classical Black–Scholes equation, while for $\mathrm{Le}>0$ the diffusion coefficient is effectively increased, leading to higher option prices that compensate for hedging costs.

Derivation of the Leland Coefficient

Assume that hedging is performed at discrete times with constant time step Δt and that each trade is subject to proportional transaction costs at rate $k>0$. Let $\Delta(t,s)=\partial_s v(t,s)$ denote the delta of the option.

Over a single hedging interval $[t,t+\Delta t]$, the change in the hedge position is

$$\Delta_{t+\Delta t}-\Delta_t\approx\frac{\partial^2 v}{\partial s^2}(t,s)\,\Delta S\quad\text{(Taylor)},$$

where $\Delta S=\sigma s\sqrt{\Delta t}\,Z$ with $Z\sim\mathcal{N}(0,1)$. Appendix A details these results.

The absolute volume of transactions is therefore

$$|\Delta_{t+\Delta t}-\Delta_t|\approx\left|\frac{\partial^2 v}{\partial s^2}(t,s)\right|\,\sigma s\sqrt{\Delta t}\,|Z|.$$

The expected transaction cost over $[t,t+\Delta t]$ is

$$\mathbb{E}[\mathrm{Cost}]=ks\,\mathbb{E}[|\Delta_{t+\Delta t}-\Delta_t|]=ks^2\sigma\sqrt{\Delta t}\left|\frac{\partial^2 v}{\partial s^2}\right|\,\mathbb{E}[|Z|].$$

Since $\mathbb{E}[|Z|] = \sqrt{\frac{2}{\pi}}$, this gives

$$\mathbb{E}[\text{Cost}] = ks^2\sigma\sqrt{\frac{2\Delta t}{\pi}}\left|\frac{\partial^2 v}{\partial s^2}\right|.$$

Dividing by the time step Δt yields an additional drift term in the pricing equation :

$$\frac{\mathbb{E}[\text{Cost}]}{\Delta t} = \frac{1}{2}\sigma^2 s^2\left(\frac{2k}{\sigma}\sqrt{\frac{2}{\pi\Delta t}}\right)\left|\frac{\partial^2 v}{\partial s^2}\right|.$$

Therefore the nonlinear diffusion adjustment appearing in the Leland PDE is obtained by introducing the *Leland number*

$$\boxed{\text{Le} = \frac{2k}{\sigma}\sqrt{\frac{2}{\pi\Delta t}}}$$

which leads to the modified diffusion coefficient

$$\frac{1}{2}\sigma^2 s^2\left(\frac{\partial^2 v}{\partial s^2} + \text{Le}\left|\frac{\partial^2 v}{\partial s^2}\right|\right).$$

This coefficient aggregates the effect of proportional transaction costs and discrete hedging into a single dimensionless constant.

Finite difference discretization of the Leland equation

We consider the Leland pricing equation

$$-\frac{\partial v}{\partial t} - \frac{1}{2}\sigma^2 s^2\left(\frac{\partial^2 v}{\partial s^2} + \text{Le}\left|\frac{\partial^2 v}{\partial s^2}\right|\right) - rs\frac{\partial v}{\partial s} + rv = 0,$$

supplemented with terminal condition $v(T, s) = \hat{g}(s)$.

Assuming convex payoffs (the case for european put option) , the solution satisfies $\frac{\partial^2 v}{\partial s^2} \geq 0$, so that the equation reduces to the linear form

$$-\frac{\partial v}{\partial t} - \frac{1}{2}\sigma^2 s^2(1 + \text{Le})\frac{\partial^2 v}{\partial s^2} - rs\frac{\partial v}{\partial s} + rv = 0.$$

Let

$$m = 1 + \text{Le}.$$

Finite difference approximations

$$s_i = ih, \quad i = 0, \dots, N, \qquad t^n = n\Delta t, \quad n = 0, \dots, M.$$

Denote $v_i^n \approx v(t^n, s_i)$.

We use a backward Euler discretization in time and centered differences in space :

$$\frac{\partial v}{\partial t}(t^{n+1}, s_i) \approx \frac{v_i^{n+1} - v_i^n}{\Delta t}, \qquad \frac{\partial v}{\partial s}(t^{n+1}, s_i) \approx \frac{v_{i+1}^{n+1} - v_{i-1}^{n+1}}{2h}, \qquad \frac{\partial^2 v}{\partial s^2}(t^{n+1}, s_i) \approx \frac{v_{i+1}^{n+1} - 2v_i^{n+1} + v_{i-1}^{n+1}}{h^2}.$$

Discrete equation

Substituting into the PDE yields :

$$\frac{v_i^{n+1} - v_i^n}{\Delta t} = m\frac{1}{2}\sigma^2 s_i^2\frac{v_{i+1}^{n+1} - 2v_i^{n+1} + v_{i-1}^{n+1}}{h^2} + rs_i\frac{v_{i+1}^{n+1} - v_{i-1}^{n+1}}{2h} - rv_i^{n+1}.$$

Multiplying by Δt and rearranging :

$$-\Delta t(m\gamma_i - \delta_i)v_{i-1}^{n+1} + (1 + \Delta t(2m\gamma_i + r))\, v_i^{n+1} - \Delta t(m\gamma_i + \delta_i)v_{i+1}^{n+1} = v_i^n,$$

where

$$\gamma_i = \frac{1}{2}\sigma^2\frac{s_i^2}{h^2}, \qquad \delta_i = \frac{rs_i}{2h}.$$

Matrix form

Collecting the coefficients gives the tridiagonal linear system

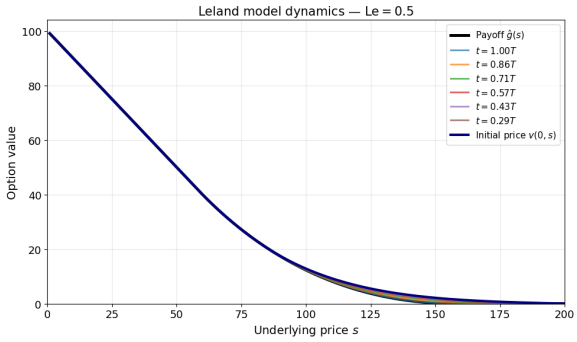
$$Av^{n+1} = v^n,$$

where the matrix A has diagonals

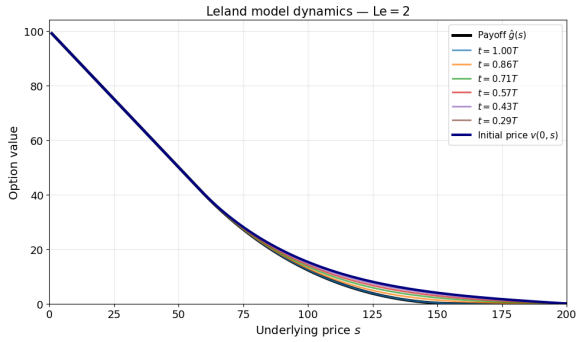
$$\begin{aligned} A_{i,i} &= 1 + \Delta t(2m\gamma_i + r), \\ A_{i,i-1} &= -\Delta t(m\gamma_i - \delta_i), \\ A_{i,i+1} &= -\Delta t(m\gamma_i + \delta_i). \end{aligned}$$

Numerical resolution

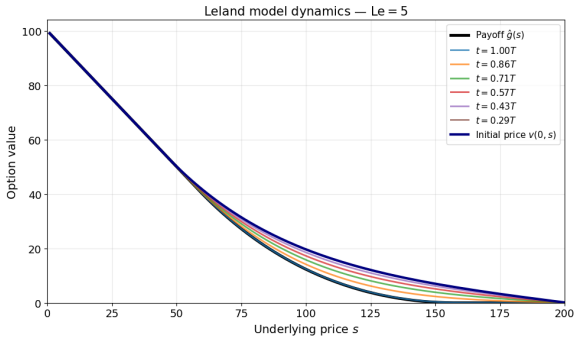
We solve the gamma-constrained Leland equation by discretizing the asset price and time using finite differences and backward Euler, then enforcing the gamma constraint as a lower obstacle on the option value. At each time step, the resulting linear complementarity problem is solved with Projected Successive Over-Relaxation (PSOR), which updates the solution while ensuring it remains above the obstacle.



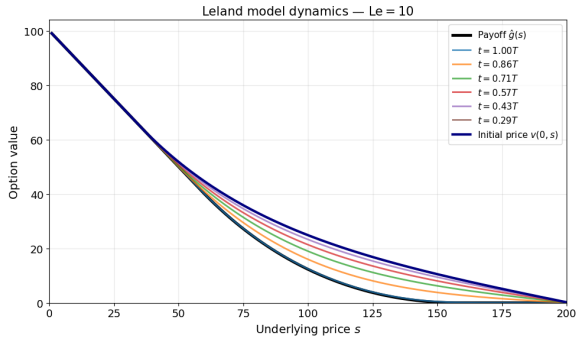
(a) Option pricing with Le = 0.5



(b) Option pricing with Le = 2



(c) Option pricing with Le = 5



(d) Option pricing with Le = 10

FIGURE 8 – Numerical solution for different Le values.

As Le increases, the option value at earlier times becomes higher because transaction costs (or leverage effects in the Leland model) make perfect hedging more expensive, which inflates the option premium. The time dynamics become stronger for large Le : the gap between the initial price and the payoff is larger, so the option value decays toward its payoff over time more noticeably (and from a higher level). Consequently, when Le is small, the pricing curves are very close to the constrained classical Black–Scholes solution, while for large Le, Leland's transaction-cost correction dominates the pricing behavior.

Appendix A : Justification of the Discrete Price Increment Approximation

In the Black–Scholes framework, the underlying asset price $(S_t)_{t\geq 0}$ follows under the risk–neutral measure the stochastic differential equation

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t,$$

where $(W_t)_{t\geq 0}$ denotes a standard Brownian motion.

This equation admits the explicit solution

$$S_{t+\Delta t} = S_t \exp\big((r - \tfrac{1}{2}\sigma^2)\Delta t + \sigma(W_{t+\Delta t} - W_t)\big)\,.$$

Let

$$\Delta W = W_{t+\Delta t} - W_t.$$

By the properties of Brownian motion,

$$\Delta W \sim \mathcal{N}(0, \Delta t),$$

hence there exists a standard normal random variable $Z \sim \mathcal{N}(0,1)$ such that

$$\Delta W = \sqrt{\Delta t}\,Z.$$

Using the Taylor expansion of the exponential function $e^x = 1 + x + O(x^2)$ as $x \to 0$, we obtain

$$S_{t+\Delta t} = S_t\Big(1 + \sigma\sqrt{\Delta t}\,Z + O(\Delta t)\Big).$$

Therefore, the price increment satisfies

$$\Delta S := S_{t+\Delta t} - S_t = \sigma S_t\sqrt{\Delta t}\,Z + O(\Delta t).$$

Since transaction costs are proportional to $|\Delta S|$ and will be divided by Δt when expressed per unit of time, the stochastic term of order $\sqrt{\Delta t}$ strictly dominates the deterministic term of order Δt. Consequently, only the Brownian component of the price increment is retained in the Leland approximation, leading to the asymptotic representation

$$\Delta S \approx \sigma S_t\sqrt{\Delta t}\,Z.$$